
Algorithm Flowchart & Pseudo-code

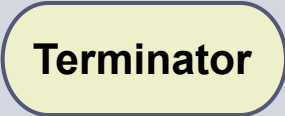
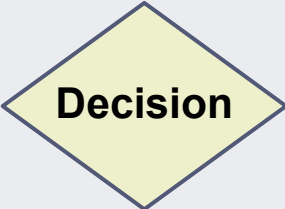
- ❑ Algorithm Flowchart
- ❑ Pseudo-code

Algorithm Flowchart

Flowchart

- ▶ A graphical representation of the sequence of operations in an algorithm
- ▶ Shows the sequence of instructions in a program
- ▶ Helps visualize and understand how a program works

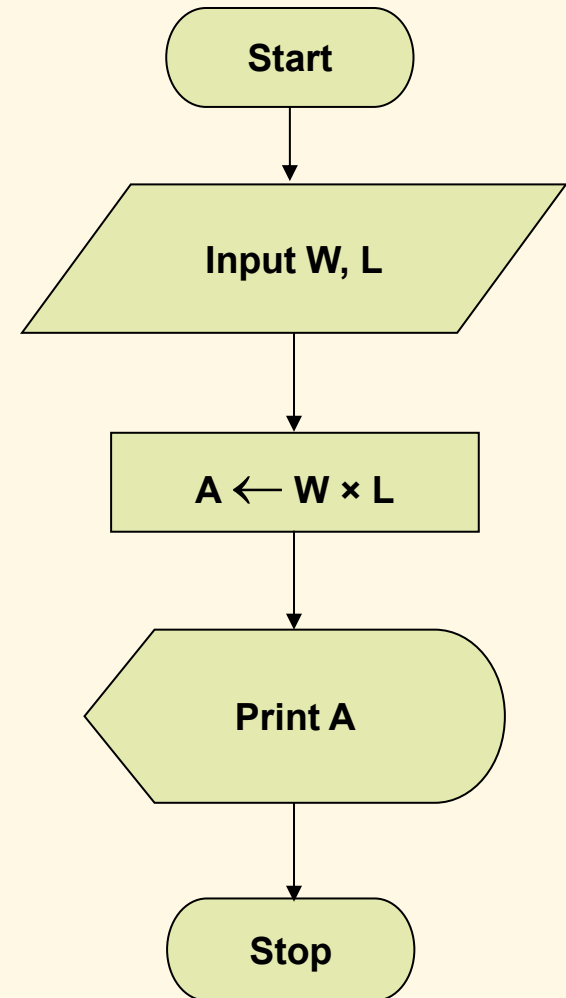
Basic Elements (*)

Element	Description
 Terminator	Beginning or end of a process
 Process	A step (task, action) in a process
 Arrow	Directional execution flow
 Decision	Conditional decision, this-or-that choice branch

Element	Description
 Data	Process input/output
 Display	Displayed output
 On-page connector	Flow continues at a target on the same chart (to avoid long arrows)
 Off-page connector	Flow continues at a target on another page

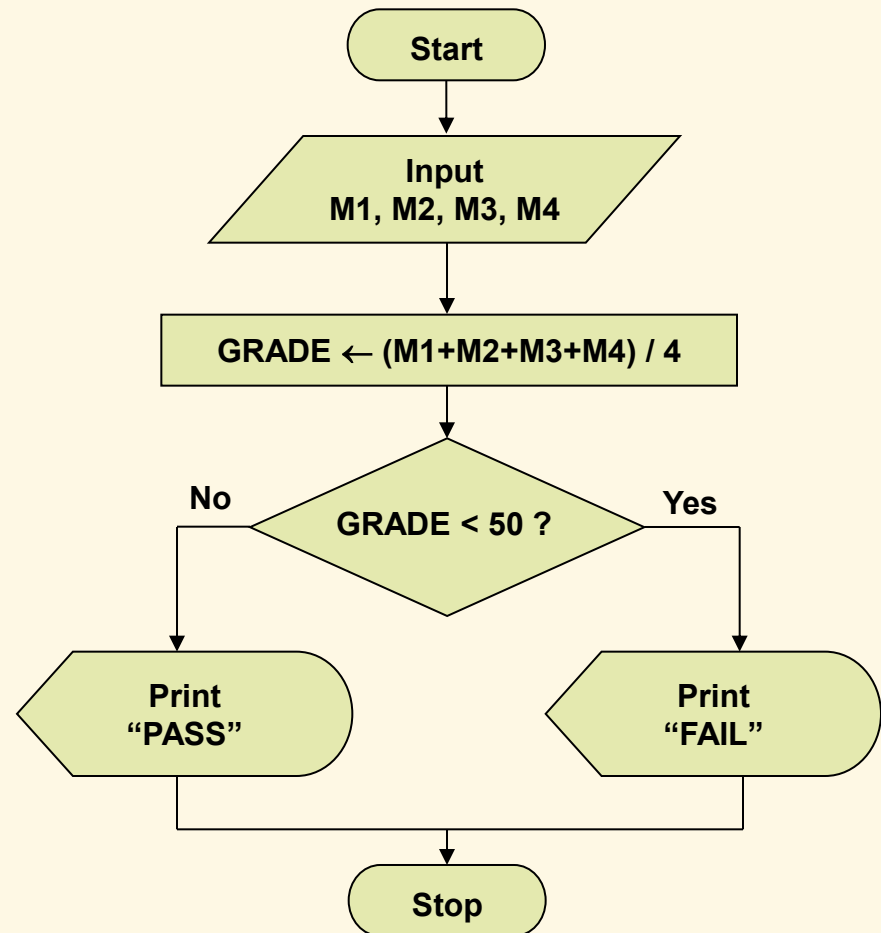
Example 1

- Write an algorithm and draw a flowchart that will read the two sides of a rectangle and calculate its area.



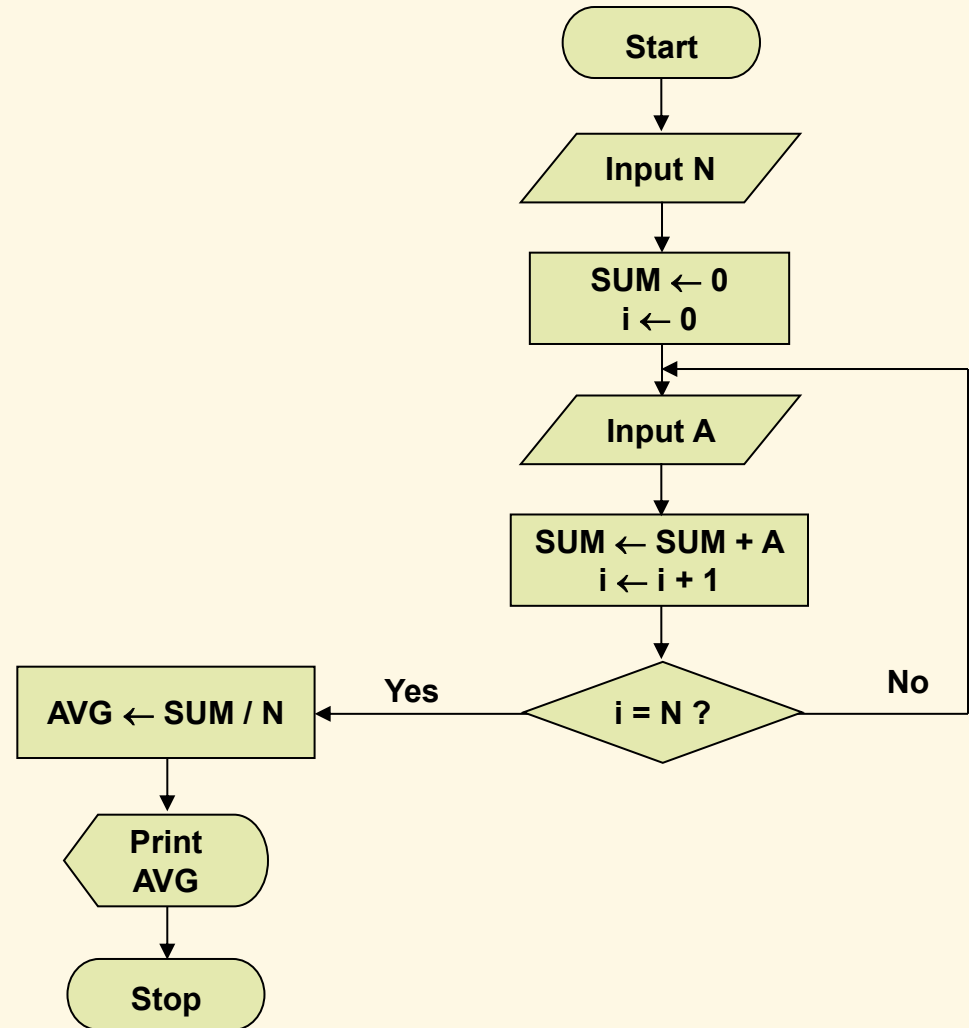
Example 2: Conditional Branch

- Write an algorithm to determine a student's final grade and indicate whether it is passing or failing. The final grade is calculated as the average of four marks.

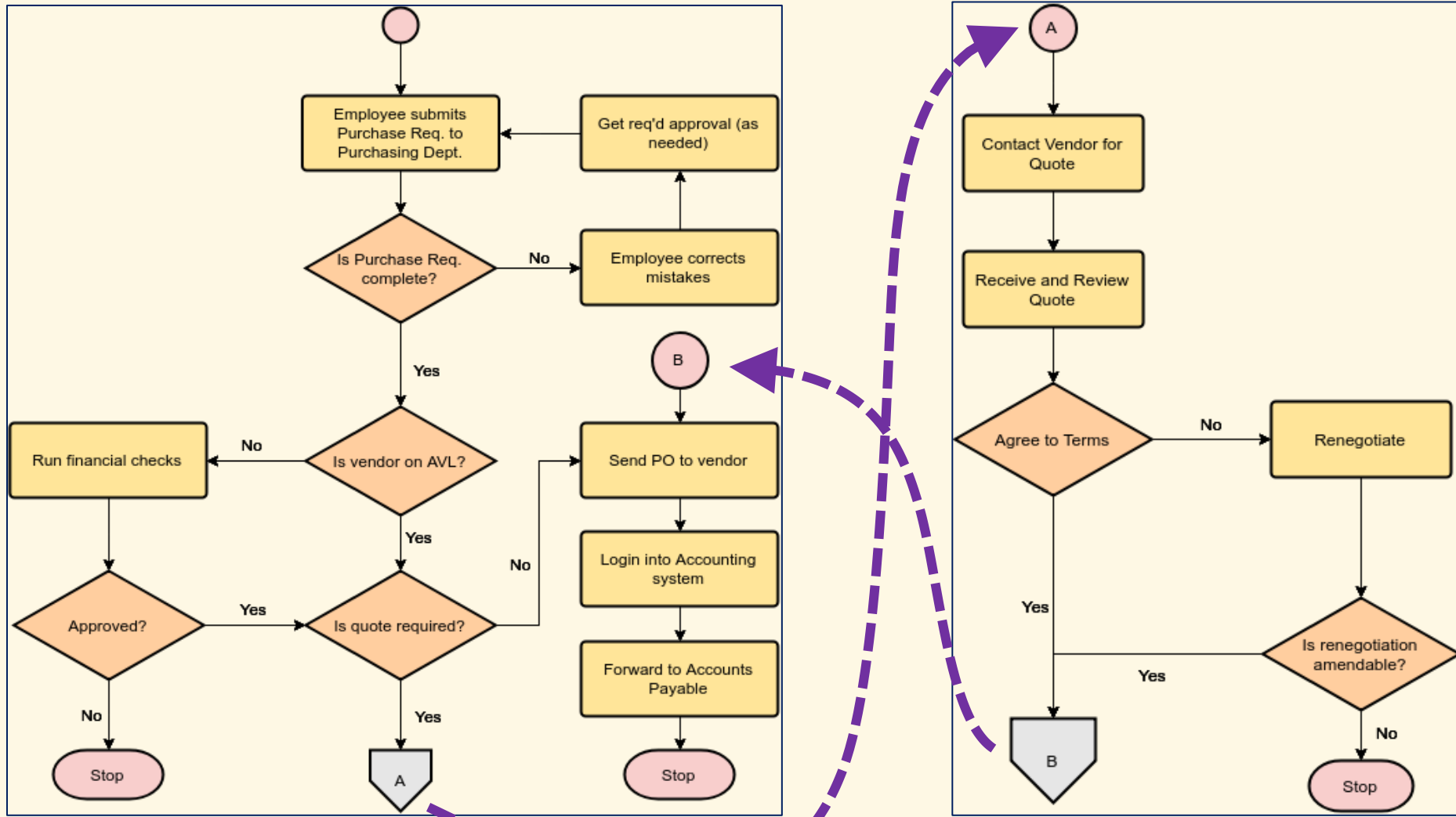


Example 3: Loop

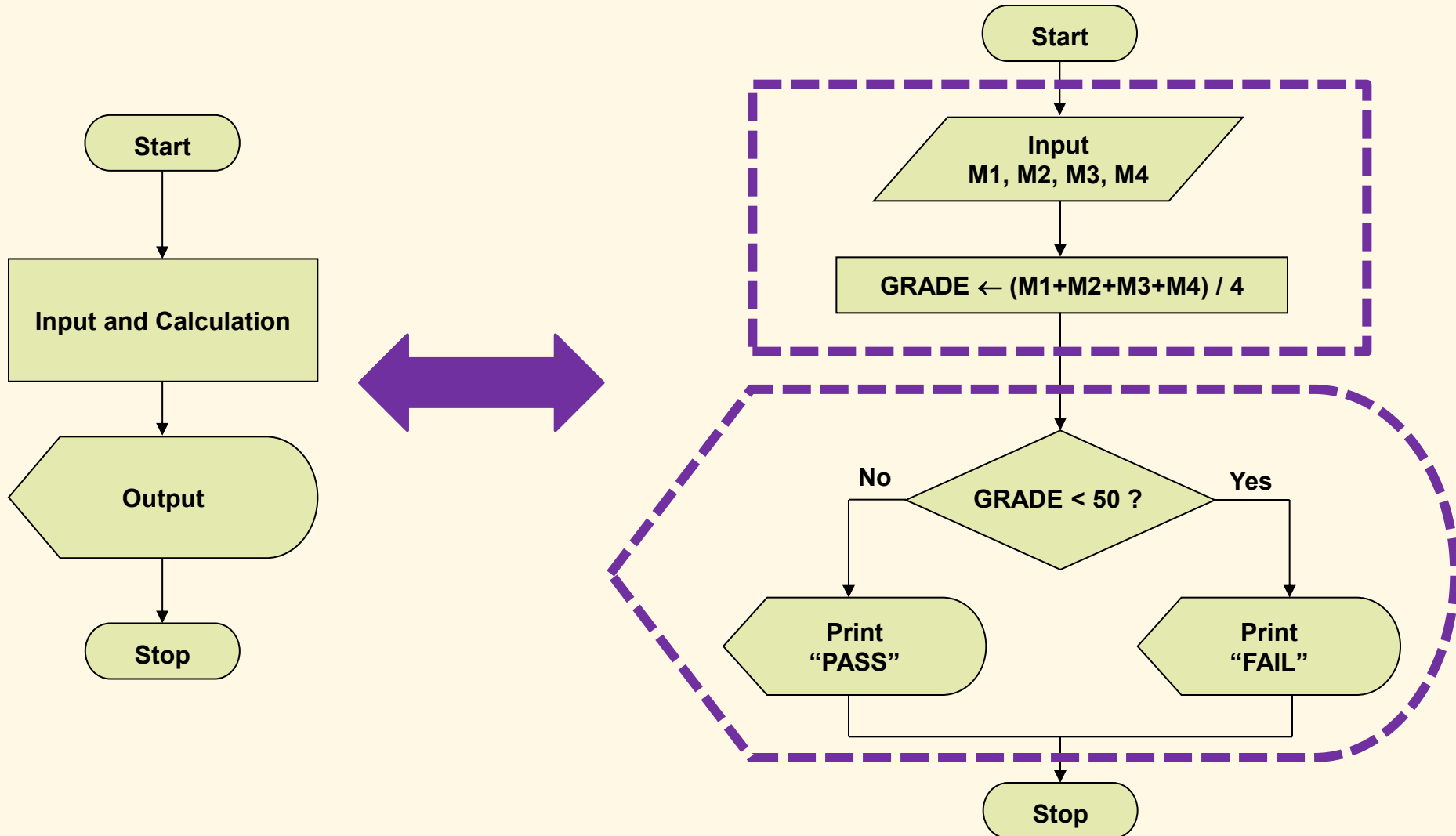
- Write an algorithm to determine the mean value of N numbers entered by user.



Linking Flowcharts



General and Detailed Flowcharts



Exercises

1. Make a flowchart to show how to solve quadratic equations: $ax^2 + bx + c = 0$.
2. Given an array of N integers, make a flowchart to print all the even numbers.
3. Make a flowchart of a ticket vending machine that accepts payment by cash and card.
4. Make a flowchart to calculate the taxi fare given the distance driven and time waited, with following rules:
 - ▶ Fare = Fare-by-distance + Fare-by-waiting-time
 - ▶ First 2km driven: \$5
 - ▶ Next every 0.1km driven: \$0.15
 - ▶ First 30min waited: \$3
 - ▶ Next every 10min waited: \$0.5

Pseudo-code



Pseudo-code

- ▶ Artificial and informal language that helps programmers develop algorithms
- ▶ May be combinations of computer language and spoken language

Common Statements

- ▶ Assignment
 - ▶ set *variable* to *value*
 - ▶ *variable* $\leftarrow$ *arithmetic-expression*
- ▶ If-then
 - ▶ if *condition* then
 statement
 [end/endif]
- ▶ If-then-else
 - ▶ if *condition* then
 statement-#1
else
 statement-#2
 [end/endif]

Common Statements (*cont'd*)

- ▶ Case-of

- ▶ case *expression* of

- condition-#1: statement-#1*

- condition-#2: statement-#2*

- ...*

- condition-#n: statement-#n*

- others: default-statement*

- [end]*

- ▶ Invoking procedures or functions

- ▶ *call proc*

- ▶ *call proc* with *parameters*

- ▶ *call func* with *parameters* returning *value*

Common Statements (*cont'd*)

▶ Loops

▶ repeat

statement

until *condition*

▶ do

statement

while *condition*

▶ while *condition* do

statement

▶ for each *element* in *list* do

statement

▶ for *variable* $\leftarrow$ *v1* [to|down to] *v2* do

statement

Procedures & Functions

- ▶ Example: Find the maximum element of an array.
- ▶ Pseudo-code
 - ▶ Algorithm `ArrayMax(A, n)`:
Input: Array A storing n integers.
Output: The maximum element in A .

```
currentMax  $\leftarrow$   $A[0]$ 
for  $i \leftarrow 1$  to  $n-1$  do
    if currentMax  $<$   $A[i]$  then
        currentMax  $\leftarrow$   $A[i]$ 

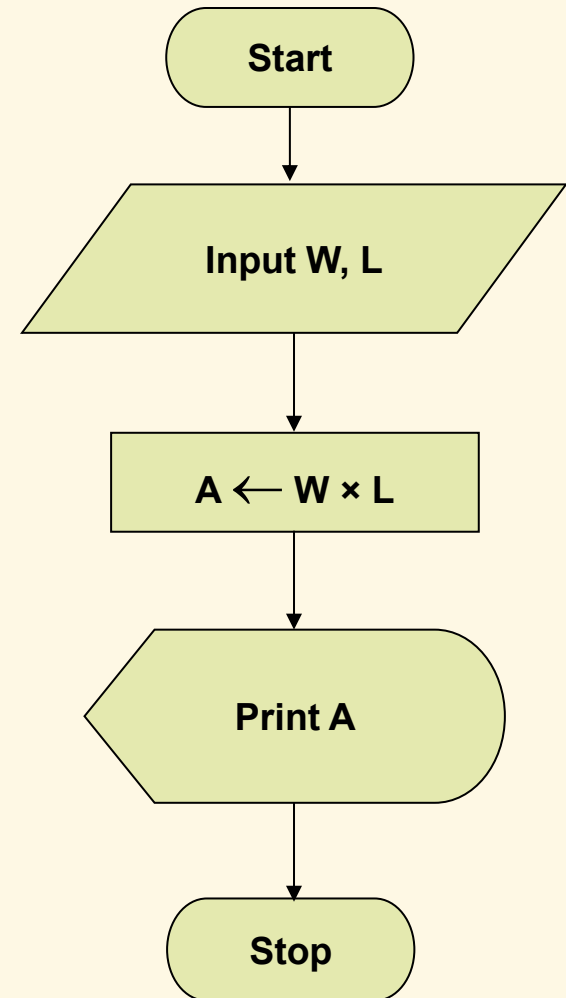
return currentMax
```


Example 1

Input W, L

$A \leftarrow W \times L$

Print A



Example 2: Conditional Branch

Input M_1, M_2, M_3, M_4

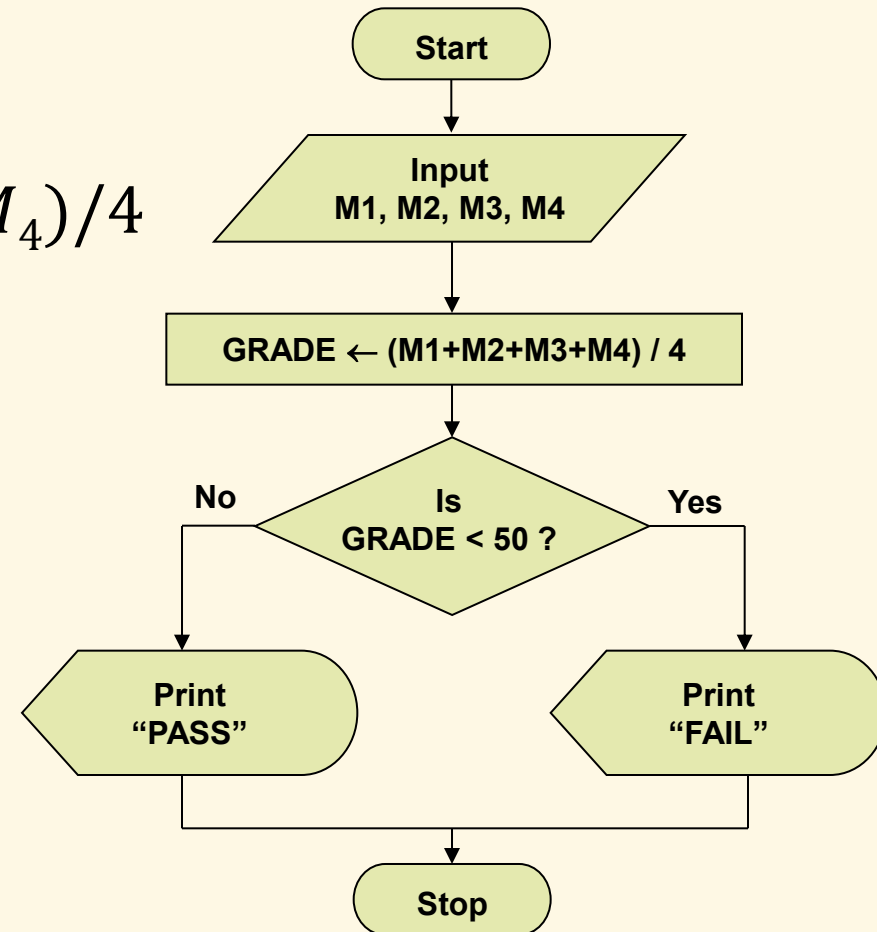
$$GRADE \leftarrow (M_1 + M_2 + M_3 + M_4) / 4$$

if $GRADE < 50$ then

 Print “FAIL”

else

 Print “PASS”



Example 3: Loop

Input N

$SUM \leftarrow 0$

$i \leftarrow 0$

repeat

 Input A

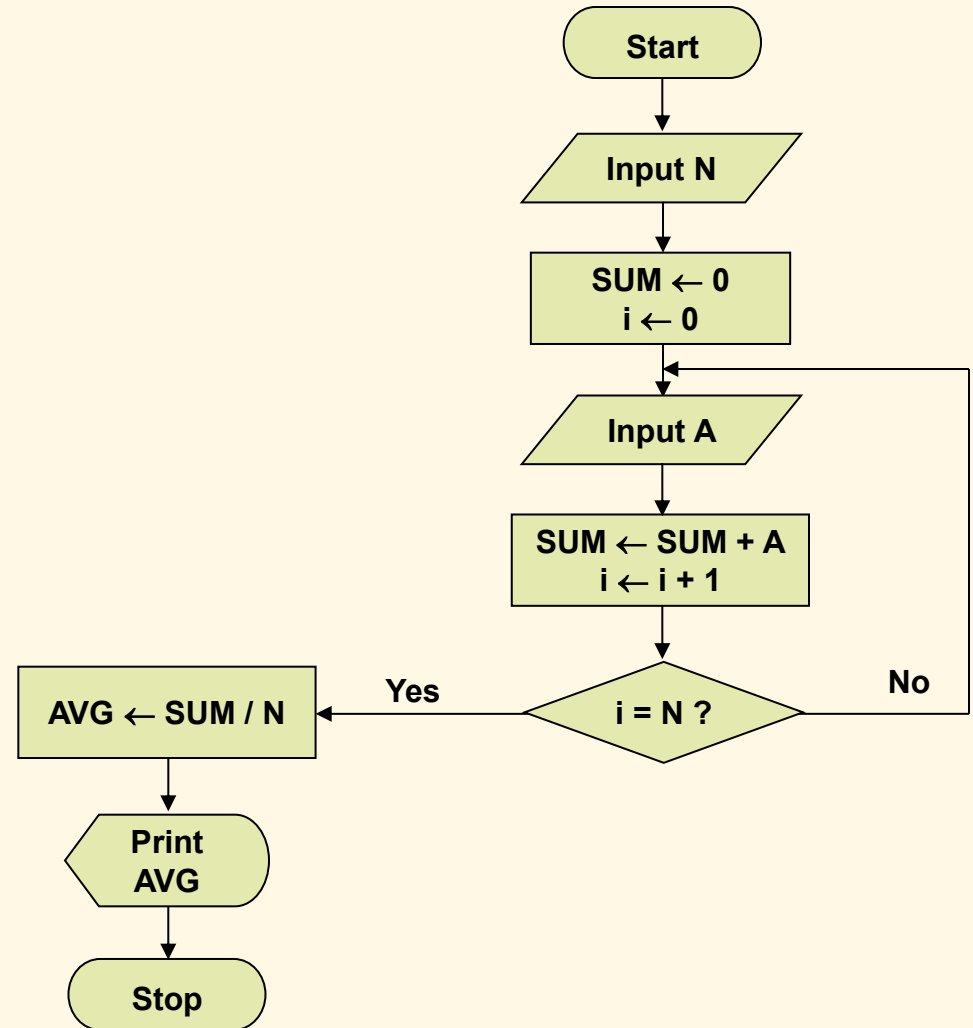
$SUM \leftarrow SUM + A$

$i \leftarrow i + 1$

until $i = N$

$AVG \leftarrow SUM / N$

Print AVG



Exercises

- ▶ Revisit the flowchart exercises and rewrite the algorithms using pseudo-code.